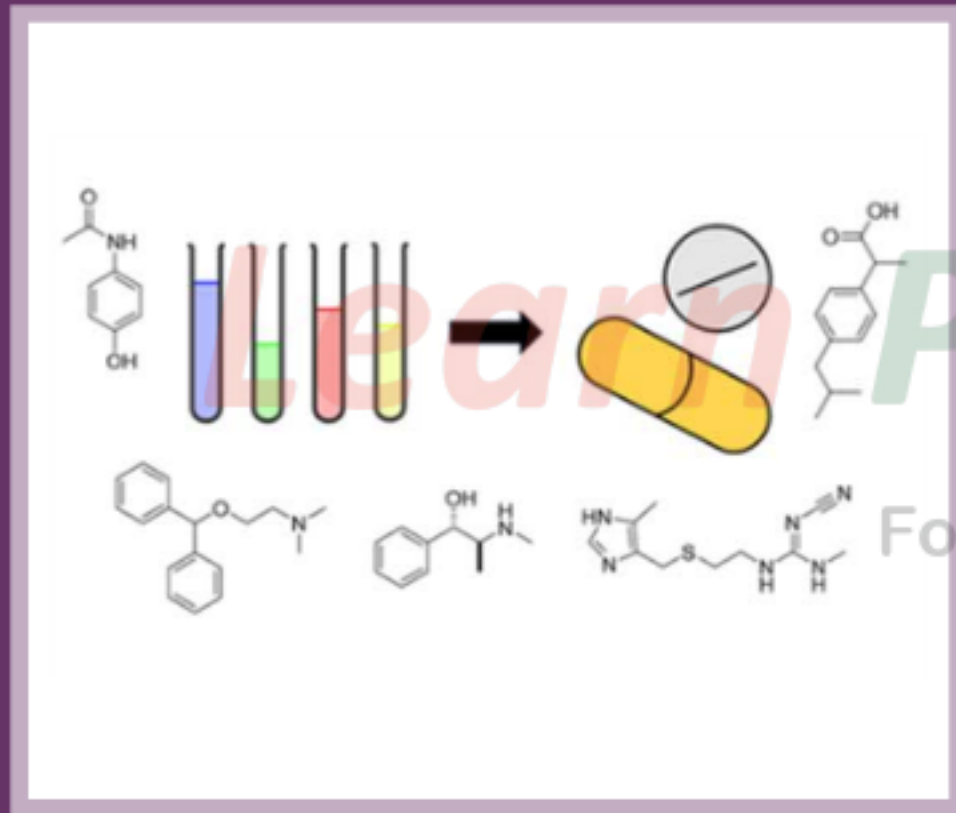


HISTORY AND DEVELOPMENT OF MEDICINAL CHEMISTRY



🎯 History and development of medicinal chemistry

For Pharmacy Courses



History And Development of Medicinal Chemistry

- The ancient civilizations of Babylon, China, Egypt, and India found that various plants were used for treating various diseases.
- The therapeutic properties of plants were described by the ancient Greeks and Romans and are documented in the writings of Hippocrates and Galen.
- The De materia medica (50-70 CE) is a pharmacopoeia of medicinal plants used to prepare medicines was written by Pedanius Dioscorides which was read for more than 1500 years.
- Herbals from John Gerard (1596), John Parkinson (1640) and Nicolas Culpeper (1649) provide insights into this widespread practice.
- As a result of explorations in the seventeenth and eighteenth centuries, a number of useful tropical plants were discovered.
- In the beginning of the nineteenth century (1803), morphine was made from the poppy plant and used for pain relief.
- By the middle of the nineteenth century (1853), Henry modified the functional group in natural products (morphine) to make quaternary ammonium salts of morphine for cough sedation.
- In the nineteenth century (known as the age of innovation and chemistry) chemicals like diethyl ether (1842), nitrous oxide (1845) and chloroform (1847) were introduced as general anaesthetics, to perform a tumour removal surgery in the neck part.
- Various antiseptics, such as iodine (1839) and phenol (1860), also played a key role in surgery's success.
- As a traditional pain reliever, willow bark was discovered to contain a yellow crystal called salicin (1828) which was then refined by Raffaele Piria in (1838) and obtained a stronger compound from the crystals named salicylic acid.
- The use of salicylic acid for typhoid fever treatment began in 1875.
- The local anaesthetic effect of cocaine was first reported in 1884, but the structure at that time was unknown. Various modifications to cocaine led to the discovery of benzocaine (1892) and procaine (1905).
- In 1890 Ehrlich proposed that biologically active compounds have specific receptors and that these compounds have a lock-and-key relationship with one another.

- In the 20th century, vitamin deficiency diseases were recognized and the structure of various vitamins was revealed. Following that, synthetic antimalarials were introduced as a replacement for quinine, such as pamaquine (1926), mepacrine (1932), and later chloroquine.
- The discovery of penicillin in 1928, the world's first antibiotic, was one of Alexander Fleming's greatest achievements.
- In 1940, nitrogen mustard was discovered as an alkylating agent for the treatment of cancer.
- Thalidomide (1950s) was a widely used drug for treatment of nausea in pregnant women resulting in severe birth defects in thousands of children.
- These consequences brought a major tightening of the regulations in terms of registration and the safety of medicines.
- At last,attaining drug selectivity is now a high priority in drug design.

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